

Lab 1 — Use-Case Flow Specification

Restaurant Table Booking & Pre-Ordering App | Problem Statement #36

PES University — Dept. of CSE • Name: _____ SRN: _____ Date: _____

Use Case ID	UC-01
Use Case Name	Book Table with Pre-Order
Scope / Level	Restaurant Table Booking & Pre-Ordering App — User-goal level
Primary Actor	Diner
Secondary Actors	Payment Gateway (external), Kitchen Display System – KDS (external), Restaurant Manager (floor terminal)
Traces to	FR-001, FR-002, FR-003, FR-004 NFR-001, NFR-002
Trigger	Diner taps “Book a Table” and selects restaurant, date, time slot and party size.

Preconditions

- P1** The Diner is registered and authenticated in the app.
P2 The restaurant's floor plan, table inventory and digital menu are published and live.
P3 The payment gateway and the Kitchen Display System integrations are reachable.

Postconditions

Success guarantee:

- Q1** A reservation exists with a unique booking ID and status CONFIRMED; the selected table is locked for the whole dining window.
Q2 The pre-order lines are linked to the booking ID and visible on the Restaurant Manager's terminal.
Q3 A kitchen preparation timeline is computed and the ticket is scheduled for release to the KDS.
Q4 Payment is authorised (only a gateway token is stored) and a digital confirmation with a QR check-in code is delivered to the Diner.
Minimal guarantee (on failure): no table remains locked, no amount is captured, no kitchen ticket is queued; the cart is preserved so the Diner can retry.

Main Success Scenario

- Diner selects the restaurant, date, time slot and party size.
- System retrieves the live floor plan and displays every table colour-coded as Available / Reserved / Seated. «include» UC-02 Validate Table Availability.
- Diner taps an available table on the 2D floor plan.
- System places a 10-minute soft hold on that table for the requested dining window and confirms the selection.
- Diner chooses to add a pre-order. «extend» UC-03 Pre-Order Menu Items (optional — the Diner may book the table alone).
- System displays the digital menu; Diner adds items with quantity and customisation notes.
- System links the cart to the pending booking and shows the itemised total.
- System computes each item's cook-start time from its prep duration and the reserved arrival time and displays the estimated serve time. «include» UC-04 Generate Kitchen Prep Timeline.
- Diner confirms the booking and proceeds to payment. «include» UC-05 Make Payment (optionally «extend» UC-06 Apply Promo Code before authorisation).
- System sends the payment request to the Payment Gateway over TLS 1.2+; the Gateway authorises it and returns a token (NFR-002).
- System converts the soft hold into a confirmed reservation, sets the table status to Reserved and stores only the payment token.
- System pushes the status change to all manager floor terminals within 500 ms (NFR-001) and schedules the kitchen ticket for the computed cook-start time.
- System issues the digital confirmation — booking ID, table number, itemised pre-order and QR check-in code — by push notification and e-mail. Use case ends successfully.

Alternate Flow A1 — Payment Declined (at step 10)

- A1.1** The Payment Gateway returns “declined”, or the authorisation request times out.
A1.2 System keeps the reservation in PENDING, retains the soft hold and the cart, displays the decline reason and offers an alternative payment method.
A1.3 Diner selects another method — the flow resumes at step 10.
A1.4 After 2 failed attempts, or when the 10-minute hold expires, the System cancels the pending booking, returns the table to Available, pushes the update to all floor terminals and notifies the Diner. No kitchen ticket is queued and no amount is captured. Use case ends in failure.

Alternate Flow A2 — Table Taken Concurrently (at step 4)

- A2.1** Another Diner confirms the same table for an overlapping window a moment earlier.
A2.2 System refuses the hold, repaints the table as Reserved and offers the nearest alternative tables for the same slot.
A2.3 Diner picks an alternative table (resume at step 4) or exits — no reservation is created.